

CasperPay Bridge

Mobile Money ↔ Multi-Chain Settlement Layer Powered by Casper

Inspiration

Africa has already solved everyday payments through **mobile money** (M-Pesa, MTN MoMo, Airtel Money), with over **400 million active wallets**. However, these systems are **siloed**: - You cannot easily send money across networks or borders - Businesses face high fees, delays, and reconciliation issues - Existing blockchains rarely integrate cleanly with real-world payment rails

CasperPay Bridge is inspired by a simple idea:

Casper should not replace mobile money — Casper should connect it.

Casper's enterprise-grade blockchain, predictable fees, and Rust smart contracts make it an ideal **neutral settlement and coordination layer** for real-world payment systems.

What It Does

CasperPay Bridge enables **interoperable mobile money transfers** by using **Casper as the escrow and settlement backbone**.

With CasperPay Bridge, a user can: - Initiate a payment from one mobile money network - Have the funds securely locked and routed via Casper - Receive the value on another network or blockchain

All coordination logic is handled by **Casper smart contracts**, while off-chain relayers interface with mobile money and other chains.

Core Features

On-Chain Escrow (Casper Smart Contracts)

- Funds are locked in a Casper escrow contract
- Each transaction has a unique route ID and metadata
- Settlement only occurs when both sides are verified

Interoperability Layer

- Casper acts as the neutral settlement hub
- Supports routing to:
- Other blockchains (e.g. Stellar, Celo – testnet)

- Mobile money networks (via simulators for MVP)

Route & Fee Optimization

- Simple routing logic selects the cheapest or fastest path
- Fees are transparently calculated and recorded on-chain

Mobile Money Simulation (Hackathon MVP)

- Realistic API simulators for M-Pesa / MTN MoMo
- Web-based UI to emulate phone confirmations
- Clearly documented as mock infrastructure

Real-Time Tracking Dashboard

- Transaction lifecycle visualization
- Casper Testnet explorer links
- Status: initiated → escrowed → routed → settled

How We Built It

1. Casper Smart Contracts (Rust)

Deployed on **Casper Testnet**: - Escrow contract - Settlement confirmation logic - Fee accounting - Route metadata storage

2. Relayer & Bridge Service

- Off-chain service (Node.js / Rust)
- Monitors Casper events
- Verifies mobile money / cross-chain actions
- Triggers settlement calls

3. Frontend Dashboard

- Built with React
- Mobile-money-inspired UI
- Transaction status and logs
- Admin panel for bridge operators

4. Testnet & Simulation

- Casper Testnet for all on-chain actions
- Mock APIs for mobile money
- Optional integration with another testnet chain

Demo Flow (5 Minutes)

1. User initiates payment

2. "Send \$20 from M-Pesa (Kenya) to MTN MoMo (Uganda)"

3. Casper Escrow Activation

4. Funds locked in Casper smart contract

5. Transaction visible on Casper explorer

6. Route Execution

7. Relay verifies simulated mobile money input

8. Cross-chain or off-chain step executed

9. Settlement & Confirmation

10. Casper releases funds

11. Recipient confirmation displayed

12. Impact Summary

13. Faster settlement

14. Transparent fees

15. Interoperable infrastructure

Why This Project Can Win

Perfect Casper Alignment

- Uses Casper as **core infrastructure**, not an add-on
- Rust smart contracts handle critical logic
- Enterprise-grade payment coordination

High Real-World Impact

- Targets 400M+ mobile money users
- Solves real cross-border and cross-network friction
- Relevant to SMEs, NGOs, and enterprises

Interoperability-Focused

- Directly qualifies for **Best Interoperability App**
- Showcases Casper's role as a settlement layer

Strong Community Voting Appeal

- Easy to understand
- Visually impressive demo
- Clear African use case

Challenges We Ran Into

- Designing secure escrow logic without real mobile money access
- Balancing realism with hackathon scope
- Simplifying cross-chain complexity for a clear demo

These were solved through **clean abstractions, simulators, and clear documentation.**

Accomplishments We're Proud Of

- Fully deployed Casper Testnet smart contracts
 - End-to-end demo of interoperable payments
 - Clear separation between on-chain trust and off-chain integrations
 - Enterprise-ready architecture with real-world relevance
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What We Learned

- Casper excels as a **coordination and settlement layer**
 - Interoperability is as much about UX as protocols
 - Real-world payment problems need pragmatic blockchain design
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What's Next

- Integrate real mobile money APIs
 - Add cryptographic proofs for off-chain verification
 - Expand to more blockchains
 - Partner with payment providers and NGOs
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Built With

- Casper Blockchain (Testnet)
 - Rust (Smart Contracts)
 - Node.js / Rust (Relayer)
 - React (Dashboard)
 - Simulated Mobile Money APIs
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Team

Built with a focus on **real-world impact, enterprise readiness, and Casper-native design.**

CasperPay Bridge shows how enterprise blockchain can quietly power everyday payments — securely, transparently, and at scale.